Mohamed Mahmoud

TECHNICAL LEAD / SENIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Technical Lead and Senior Software Engineer with 7+ years building production systems across backend, web and mobile. Since 2022 I have designed and led Tripklik's travel booking platform: a Python/FastAPI supplier-aggregation layer, a Laravel booking and B2B finance core, and a single Next.js + TypeScript codebase serving both the consumer site and the travel-agency portal. The platform serves 1M+ users a month and 5,000+ searches and orders. I own its performance (middleware bundle −62%, main bundle −14%, result filters 7× faster), releases, CI/CD and the authentication layer. Recently connected the booking engine to AI assistants (Claude, ChatGPT) through the Model Context Protocol. Before that, led backend squads delivering telecom value-added services for operators in KSA, UAE and Kuwait.

CORE COMPETENCIES

System Design & Architecture | Next.js / React / TypeScript | Laravel & FastAPI APIs | Third-party Supplier Integration | Payments & B2B Ledgers | Redis, Queues & Event-Driven Design | Web Performance | Technical Leadership & Releases

WORK EXPERIENCE

Tripklik

Oct 2022 - Present

Technical Lead

Senior Software Engineer Oct 2022 - Oct 2024; Technical Lead since Oct 2024.
Riyadh, KSA (Remote)

- Architected a polyglot travel booking platform serving 1M+ users a month and 5,000+ searches and orders: FastAPI (Python) for high-concurrency flight and hotel search aggregation, Laravel (PHP) for transactional booking and B2B financial ledgers, Next.js + TypeScript + Redux Toolkit for the B2C site and B2B agency portal from one codebase.
- Built the supplier integration engine with a Factory pattern and a unified schema mapper, normalizing payloads from multiple third-party suppliers (Amadeus, Hotelbeds, Flyadeal).
- Protected the platform from supplier outages with a resilience layer: circuit breakers, provider-aware adaptive throttling and exponential backoff.
- Designed the B2B finance core in Laravel: multi-currency conversion (supplier to base to client), agency wallet ("pocket") management on ACID MySQL ledgers, and a payment-gateway factory (Stripe, PayPal, B2B wallets).
- Bridged Python services and the PHP booking engine with a shared Redis state machine; moved notifications and auto-booking onto event-driven background queues so they never block the user journey.
- Cut First Load JS from 653 to 572 KB (−12%), the main app bundle from 567 to 485 KB (−14%) and the middleware bundle from 94.6 to 36.2 KB (−62%); made flight-result filters 7× faster with Map indexes and virtualized the results list.
- Built the edge-middleware authentication and permission gate (route-to-permission map, OTP enforcement) and scoped single-purpose session tokens for deep links coming from AI assistants.
- Connected the booking engine to Claude and ChatGPT through the Model Context Protocol (MCP): live flight and hotel search with agency pricing, and deep links back into the portal to book.
- Led a multi-brand frontend deployed from one branch on Vercel: subdomain routing in edge middleware, environment-driven CSS injection from S3, centralized Axios interceptors for auth and headers.
- Shipped B2B order operations: void with fee preview, refunds, PNR retrieval and import, hold orders, invoices, balances, top-up requests and role-based permissions.
- Led the modernization of the multi-brand mobile apps (Tripklik, Destinia): React Native 0.85 + Expo SDK 56 with EAS multi-brand builds, per-brand runtime configuration, Kotlin and Swift build fixes, and a 645-line developer handoff guide.
- Set up AI-assisted engineering for the team: agent guides, reusable skills and safety rules derived from real incidents, speeding up delivery while keeping review quality.

Ivas

Feb 2019 - May 2022

Senior Software Engineer

Cairo, Egypt

- Directed core backend development and infrastructure for Ringback Tone delivery platforms run by major telecom operators in KSA, UAE and Kuwait.
- Moved production backends from monolithic MVC to a clean layered architecture on Laravel and MongoDB to simplify operator integrations.
- Led backend squads on API security, performance profiling and maintainable delivery.

Grand

Jun 2018 - Oct 2018

Back-End Developer

Cairo, Egypt

- Built REST APIs for mobile and web apps with a focus on performance and security.
- Defined API contracts with frontend teams.

LoadServ For Integrated Business Solution

Jan 2018 - May 2018

Full-Stack Developer

Cairo, Egypt

- Delivered frontend and backend features on team projects; took part in code reviews.

EDUCATION

BSc Computer Science Helwan University

2014 - 2018

Cairo, Egypt

Graduation project: face-recognition platform for finding missing children (Python, Flask, OpenCV, Firebase).

SKILLS

Languages: TypeScript, JavaScript, PHP 8, Python, SQL

Backend: Laravel 8-10, FastAPI, Node.js, Express, Socket.IO, REST APIs, Webhooks, JWT, Sanctum, Swagger/OpenAPI

Frontend: React, Next.js, Redux Toolkit, RTK Query, Tailwind CSS, SCSS, Storybook, Vue 3, RTL / i18n

Mobile: React Native, Expo / EAS, Swift native modules, multi-brand builds **Data:** MySQL, Redis, MongoDB, Meilisearch, Supabase

Infrastructure: Docker, Nginx, AWS (S3), Cloudflare, Vercel, CI/CD

Architecture: Microservices, event-driven architecture, multi-tenant / multi-brand SaaS, resilience patterns (circuit breaker, backoff), design patterns

AI: Model Context Protocol (MCP), Claude / ChatGPT integrations, AI coding agents, n8n